# Windows Troubleshooting Lab

## 1. Project Overview

**Objective:** Practice basic system diagnostics and document troubleshooting procedures on Windows 11 Pro 23H2.
**Environment:** Dell Latitude 5420 — Windows 11 Pro 23H2

## 2. Test Record

### Test 01 — System Identification

**Operating system: Windows 11 Pro 23H2**
**Command used: *systeminfo***
**Expected observation: *"systeminfo"* displays the operating system, system manufacturer, system model, processor, and memory information.**
**Actual observation: The command completed successfully and displayed the Windows edition, system model, processor, and memory details.**
**Result:** Pass

### Test 02 — Network Loopback

**Operating system: Windows 11 Pro 23H2**
**Command used: *ping 127.0.0.1***
**Expected observation: The local TCP/IP loopback address responds successfully.**
**Actual observation: Four packets were sent, four replies were received, and packet loss was 0%.**
**Result:** Pass

### Test 03 — Disk Space

**Operating system: Windows 11 Pro 23H2**
**Command used: *Get-Volume***
**Expected observation: Volume information displays without errors.**
**Actual observation: *"Get-Volume"* displayed the C: volume as *NTFS*, w/ a total size of *235.8 GB* and *160.87 GB* remaining. The volume reported Healthy status and operational status OK.**
**Result:** Pass

### Test 04 — Memory Usage

**Operating system: Windows 11 Pro 23H2**
**Command used:**
```
Get-CimInstance Win32_OperatingSystem |
Select-Object `
@{Name="Total RAM
(GB)";Expression={[math]::Round($_.TotalVisibleMemorySize/1MB,2)}},
@{Name="Free RAM (GB)";Expression={[math]::Round($_.FreePhysicalMemory/1MB,2)}}
@{Name="Used RAM (GB)";Expression={[math]::Round(($_.TotalVisibleMemorySize-
$_.FreePhysicalMemory/1MB,2)}}
```
**Expected observation: Total, free, and calculated used RAM appear in GB.**
**Actual observation: The command reported 15.39 GB total visible RAM, and 7.64 GB calculated used RAM at the time of testing.**
**Result:** Pass

### Test 05 — Network Configuration

**Operating system: Windows 11 Pro 23H2**
**Command used: *ipconfig /all***
**Expected observation: The command displays network adapter configuration, including IP addressing, subnet mask, gateway, and DNS information where applicable.**
**Actual observation: "*ipconfig /all*" displayed the Wi-Fi adapter configuration, including DHCP status, IP addressing, subnet mask, default gateway, and DNS server information. Sensitive values were redacted in the evidence screenshot.**
**Result:** Pass

# 3. Findings

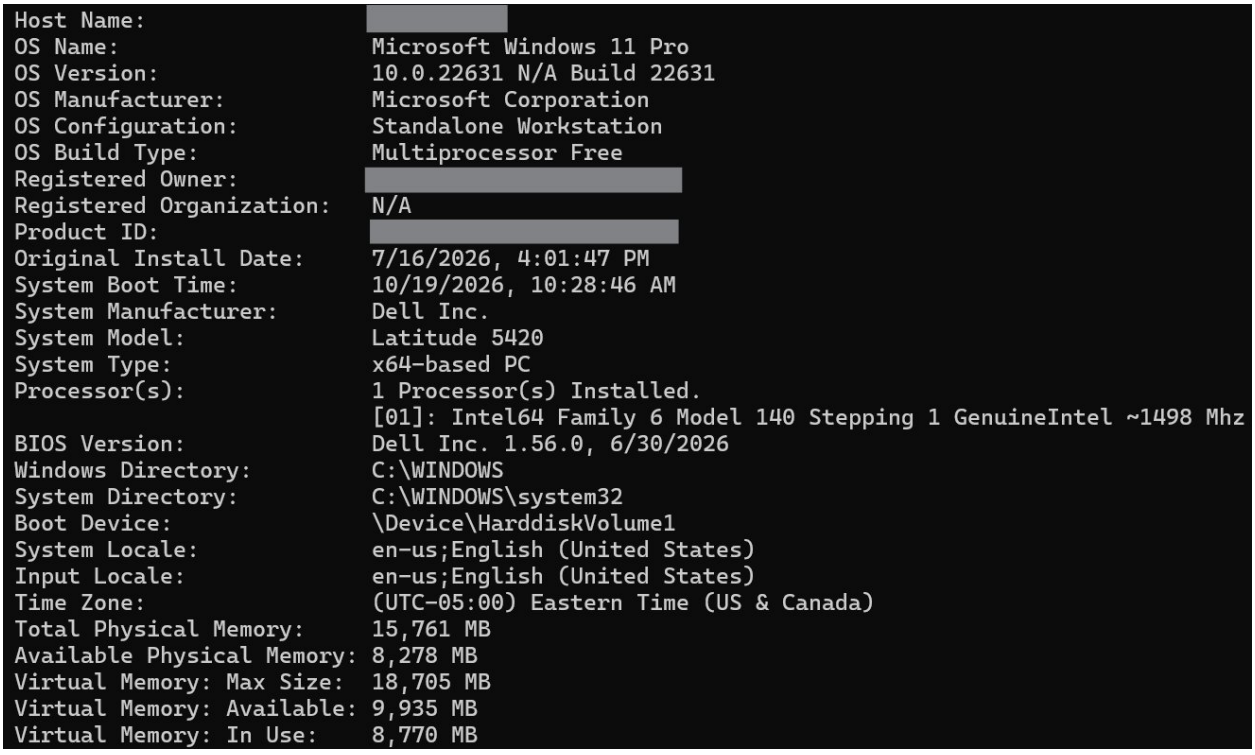
The initial diagnostic tests returned system information, disk volume details, memory information, and network configuration. The loopback test recorded four successful replies with 0% packet loss. No conclusions about external internet connectivity were drawn from the loopback test alone. Further testing is required to verify external network connectivity. The disk-space and memory measurements provide a baseline for future troubleshooting, but additional context is needed to determine whether available resources are sufficient for the computer's workload.

# 4. Lessons Learned

I practiced using Command Prompt and PowerShell to collect system information, test the local TCP/IP stack, inspect disk volumes, review physical memory, and examine network adapter configuration. I learned to distinguish between a successful local loopback test and a test of external network connectivity. I also learned to compare expected behavior against actual output before assigning a test result.

# 5. Evidence

***System Information***



### *Network Loopback*

```
C:\Users\smith>ping 127.0.0.1

Pinging 127.0.0.1 with 32 bytes of data:
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128

Ping statistics for 127.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

### *Disk Space*

```
PS C:\Users\smith> Get-Volume

DriveLetter FriendlyName FileSystemType DriveType HealthStatus OperationalStatus SizeRemaining     Size
----------- ------------ -------------- --------- ------------ ----------------- -------------     ----
C           OS           NTFS           Fixed     Healthy      OK                    160.87 GB 235.8 GB
            WINRETOOLS   NTFS           Fixed     Healthy      OK                    265.22 MB   990 MB
            DELLSUPPORT  NTFS           Fixed     Healthy      OK                    314.12 MB  1.38 GB
```

### *Memory Usage*

```
PS C:\Users\smith> Get-CimInstance Win32_OperatingSystem |
>> Select-Object `
>> @{Name="Total RAM (GB)";Expression={[math]::Round($_.TotalVisibleMemorySize/1MB,2)}},
>> @{Name="Free RAM (GB)";Expression={[math]::Round($_.FreePhysicalMemory/1MB,2)}},
>> @{Name="Used RAM (GB)";Expression={[math]::Round(($_.TotalVisibleMemorySize-$_.FreePhysicalMemory)/1MB,2)}}

Total RAM (GB) Free RAM (GB) Used RAM (GB)
-------------- ------------- -------------
         15.39          7.75          7.64
```

***Network Configuration***

